Master of Business Analytics

Prescriptive Analytics & Optimization

MBAN 5540

Week 08: Decision Analysis

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Decision Analysis

- We live in a world surrounded by uncertainties and risks.
- Decision Analysis is an analytic and systematic approach to the study of decision making.
- Managers have to make a decision, without knowing for sure which events will occur in the future.
- We better consider all decision alternatives before we make a decision.

The Decision Process in Operations

Define	Clearly define the problems and the factors that influence it
Develop	Develop specific and measurable objectives
Model	Develop a model
Evaluate	Evaluate each alternative solution
Select	Select the best alternative
Implement	Implement the decision and set a timetable for completion

Decision Making Terms

Alternative	a course of action or strategy that may be chosen by the decision maker
State of nature (Outcome)	an occurrence or a situation over which the decision maker has little or no control

Decision-Making Environments

Decision making under certainty	State of nature is known
Decision making under uncertainty	Complete uncertainty as to which state of nature may occur
Decision making under risk	Several states of nature may occur Each has a probability of occurring

Decision Making Under Uncertainty

- The manager can list the possible future outcomes but cannot estimate the probability that a specific outcome will occur.

Five different decision-making criteria can be used to make a decision under uncertainty:

1. Maximax
2. Maximin
3. Realism
4. Equally likely
5. Minimax regret

Pain Killer Example

- A pharmaceutical company is considering the introduction of a new pain killer. The manager would like to consider two alternatives:
  1. Develop a new pain killer
  2. Keep the existing one

- The possible outcomes (state of nature) are:
  - Very effective
  - Not effective

- The payoffs are:
  - If the new pain killer is effective: profit of $700
  - If the new pain killer is not effective: cost of $100
  - If the company keep the existing one: profit of $90

Pain Killer Example

- Payoff Table:

Alternatives		Outcomes (State of nature)	
		Effective	Not effective
1	Development the new pain killer	$700	-$100
2	Keep the existing one	$90	$90

1 - Maximax

- For each alternative, find the maximum payoff over all possible outcomes. Next, find the maximum of them. Chose the alternative whose maximum payoff gives this maximum.

- It is choosing the alternative with best payoff, if the best outcome happens (optimistic!)

	Alternatives	Outcomes (State of nature)		Max
		Effective	Not effective	
1	Development the new pain killer	$700	-$100	700
2	Keep the existing one	$90	$90	90

Max

2 - Maximin

- For each alternative, find the minimum payoff over all possible outcomes. Next, find the maximum of them. Chose the alternative whose minimum payoff gives this maximum.

- It is choosing the alternative with best payoff, if the worst outcome happens (pessimistic!)

Alternatives		Outcomes (State of nature)		Min
		Effective	Not effective	
1	Development the new pain killer	$700	-$100	-100
2	Keep the existing one	$90	$90	90

Max

3 - Criterion of Realism

- Decision makers should not be optimists or pessimists. They should be realistic.

- Define Coefficient of Realism:

$$0 \leq \alpha \leq 1$$

  - $\alpha = 1$ means the decision maker is totally optimistic (Maximax).
  - $\alpha = 0$ means the decision maker is totally pessimistic (Maximin).

- Calculate the realism payoff for each alternative and select the alternative with the highest realism payoff.

$$\text{Realism Payoff} = \alpha(\text{Max payoff}) + (1 - \alpha)(\text{Min payoff})$$

3 - Criterion of Realism

- Suppose $\alpha = 0.55$

- Alternative 1: $0.55 \times 700 + 0.45 \times (-100) = 340$
- Alternative 2: $0.55 \times 90 + 0.45 \times 90 = 90$

Alternatives		Outcomes (State of nature)		Criterion for Realism
		Effective	Not effective	
1	Development the new pain killer	$700	-$100	340
2	Keep the existing one	$90	$90	90

Max

3 - Criterion of Realism

- Suppose $\alpha = 0.15$

- Alternative 1: $0.15(700) + 0.85(-100) = 20$
- Alternative 2: $0.15(90) + 0.85(90) = 90$

Alternatives		Outcomes (State of nature)		Criterion for Realism
		Effective	Not effective	
1	Development the new pain killer	$700	-$100	20
2	Keep the existing one	$90	$90	90

Max

4 - Equally Likely (Laplace)

- When the decision maker cannot estimate the probability that a specific outcome will occur.

- Calculate the average payoff for each alternative and select the alternative with the highest average payoff.

$$\text{Average Payoff} = \frac{\sum \text{payoffs}}{n}$$

where n is the number of outcomes for alternative.

4 - Equally Likely (Laplace)

- Alternative 1: $\frac{700+(-100)}{2} = 300$

- Alternative 2: $\frac{90+90}{2} = 90$

	Alternatives	Outcomes (State of nature)		Laplace
		Effective	Not effective	
1	Development the new pain killer	$700	-$100	300
2	Keep the existing one	$90	$90	90

Max

5 - Minimax Regret

- Opportunity loss or regret
  - Is the amount lost by not picking the best alternative.

- Minimax regret finds the alternative that minimizes the maximum regret for each alternative.

- First find the regrets for all outcome-alternative combinations by:

  Best payoff for outcomes – actual payoff

- Next, find the maximum regret for alternatives and choose the minimum one among them.

5 - Minimax Regret

	Alternatives	Regret of Outcomes		Max Regret
		Effective	Not effective	
1	Development the new pain killer	700-700 = 0	90-(-100) = 190	190
2	Keep the existing one	700-90 = 610	90-90 = 0	610

Min

If the pain killer is effective, the best we can do is 700

If the pain killer is not effective, the best we can do is 90

Decision Making under Uncertainty (Summary)

1. Maximax
   - Max of maximum payoff over all possible outcomes
2. Maximin
   - Max of the minimum payoff over all possible outcomes
3. Criterion of Realism
   - Max of Realism Payoff $= \alpha \times \text{Max payoff} + (1 - \alpha) \times \text{Min payoff}$
4. Equally likely (Laplace)
   - Max of Average Payoffs
5. Minimax Regret
   - Min of maximum regret for alternatives

Decision Making Under Risk

- Used when the decision maker has some idea about the probabilities of outcomes.

- We use probabilities of outcomes to find the best alternative.

1. Expected Monetary Value (EMV)
2. Expected Opportunity Loss (EOL)
3. Expected Value of Perfect Information (EVPI)

Expected Monetary Value (EMV)

- EMV is weighted average of all possible payoffs, where weights are probabilities of outcomes

- EMV for each alternative is

$$\sum_{\text{Outcomes}} \text{Probability} \times \text{Payoff}$$

- We would like to maximize the EMV

- Select the alternative with largest EMV

Expected Monetary Value (EMV)

$$\text{EMV } 1 = 0.3(200{,}000) + 0.5(100{,}000) + 0.2(-120{,}000) = 86{,}000$$

$$\text{EMV } 2 = 0.3(90{,}000) + 0.5(50{,}000) + 0.2(-20{,}000) = 48{,}000$$

Alternatives		Outcomes			EMV
		High Demand	Moderate Demand	Low Demand	
1	Build Large Clinic	200,000	100,000	-120,000	86,000
2	Build Small Clinic	90,000	50,000	-20,000	48,000
3	Do Nothing	0	0	0	0
Prior Probability		0.3	0.5	0.2	

Expected Opportunity Loss (EOL)

- Opportunity loss (Regret) is the difference between the optimal payoff and the actual payoff.
- EOL is the weighted (with probabilities) average of all regrets.
- EOL is the expected cost of not picking the best solution.
- EOL for each alternative is

$$\sum_{\text{Outcomes}} \text{Probability} \times \text{Regret}$$

- We would like to minimize the EOL
- Select the alternative with smallest EOL

Expected Opportunity Loss (EOL)

$$\text{EOL } 1 = 0.3(200{,}000 - 200{,}000) + 0.5(100{,}000 - 100{,}000) + 0.2(0 + 120{,}000) = 24{,}000$$

$$\text{EOL } 2 = 0.3(200{,}000\text{-}90{,}000) + 0.5(100{,}000 - 50{,}000) + 0.2(0 + 20{,}000) = 62{,}000$$

$$\text{EOL } 3 = 0.3(200{,}000 - 0) + 0.5(100{,}000 - 0) + 0.2\,(0 - 0) = 110{,}000$$

Alternatives		Outcomes			EOL
		High Demand	Moderate Demand	Low Demand	
1	Build Large Clinic	200,000	100,000	-120,000	24,000
2	Build Small Clinic	90,000	50,000	-20,000	62,000
3	Do Nothing	0	0	0	110,000
Prior Probability		0.3	0.5	0.2	

EMV Vs. EOL

- In EMV approach, we choose the alternative with Maximum EMV.
- In EOL approach, we choose the alternative with Minimum EOL.
- In fact, these two always suggest the same alternative:
  - The alternative suggested by Max EMV is ALWAYS the same as the alternative suggested by Min EOL

Expected Value of Perfect Information (EVPI)

- A company claims that they can predict what happens in the future with certainty.
- They have the Perfect Information, and they want to sell it to us !
- We want to know the monetary value of this information.
- If we have perfect information, we select alternative with the best payoff.
- Expected Value with Perfect Information (EVwPI)
  - Weighted average of payoffs with certainty

$$\sum_{Outcomes} \text{Probability} \times \text{Best Payoff}$$

Expected Value of Perfect Information (EVPI)

- EVwPI is the average payoff when we have the perfect information.
- Max EMV (Min EOL) is what we can do without any information ! (expected value without perfect information)
- Therefore, EVwPI minus Max EMV is the maximum worth of perfect information:

$$\text{EVPI} = \text{EVwPI} - \text{Max EMV}$$

- We must not pay more than EVPI for the information.
- EVPI is the upper bound for the money that we should spend for the information.

$$\text{EVPI} = \text{Min EOL}$$

Expected Value of Perfect Information (EVPI)

$$\text{EVwPI} = 0.3(200{,}000) + 0.5(100{,}000) + 0.2(0) = 110{,}000$$

$$\text{EVPI} = \text{EVwPI- Max EMV} = 110{,}000 - 86000 = 24{,}000$$

$$\text{EVPI} = \text{Min EOL} = 24{,}000$$

	Alternatives	Outcomes			EMV	EOL
		High Demand	Moderate Demand	Low Demand		
1	Build Large Clinic	200,000	100,000	-120,000	86,000	24,000
2	Build Small Clinic	90,000	50,000	-20,000	48,000	62,000
3	Do Nothing	0	0	0	0	110,000
	Prior Probability	0.3	0.5	0.2		

Expected Value of Perfect Information (EVPI)

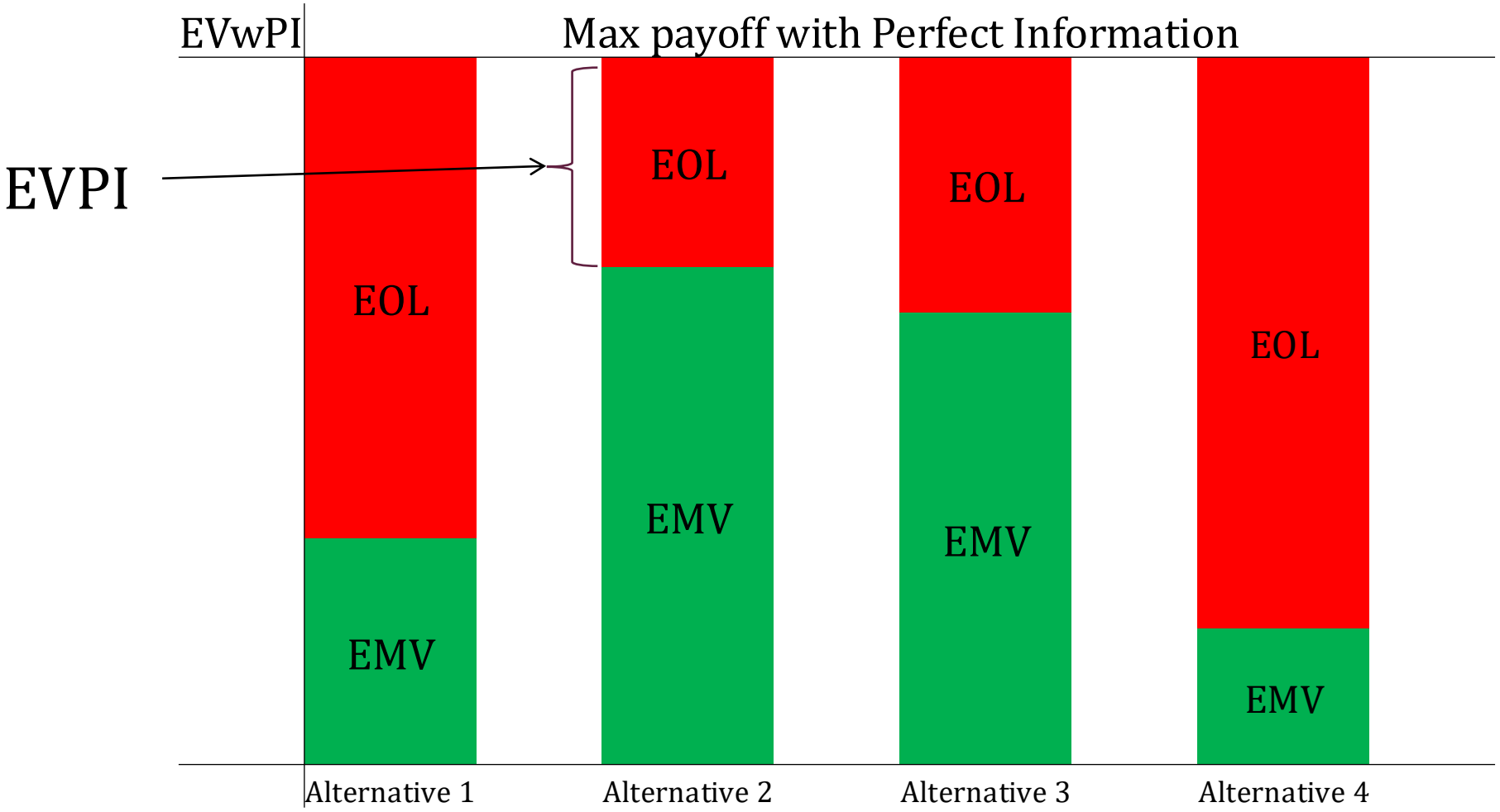
We have

- EVPI = EVwPI – Max EMV
- EVPI = Min EOL

Therefore: EVwPI = EMV + EOL

Alternatives		Outcomes			EMV	EOL	EVwPI
		High Demand	Moderate Demand	Low Demand			
1	Build Large Clinic	200,000	100,000	-120,000	86,000	24,000	110,000
2	Build Small Clinic	90,000	50,000	-20,000	48,000	62,000	110,000
3	Do Nothing	0	0	0	0	110,000	110,000

Expected Value of Perfect Information (EVPI)



Expected Value of Sample Information

- Sometimes, market research companies might offer to sell us information using surveys.

- What is the actual value of survey (sample) information?

- EVSI will determine it:
  - EVSI = EMV of the best decision with sample information (when its cost is zero) *minus* EMV of the best decision without any information

Efficiency of Sample Information

- Should we immediately accept the offer if the EVSI is greater than the cost of information?

- We should compare the EVPI and EVSI

$$\text{Efficiency} = \frac{EVSI}{EVPI}$$

- We should go with offers with higher efficiency.

Decision Trees

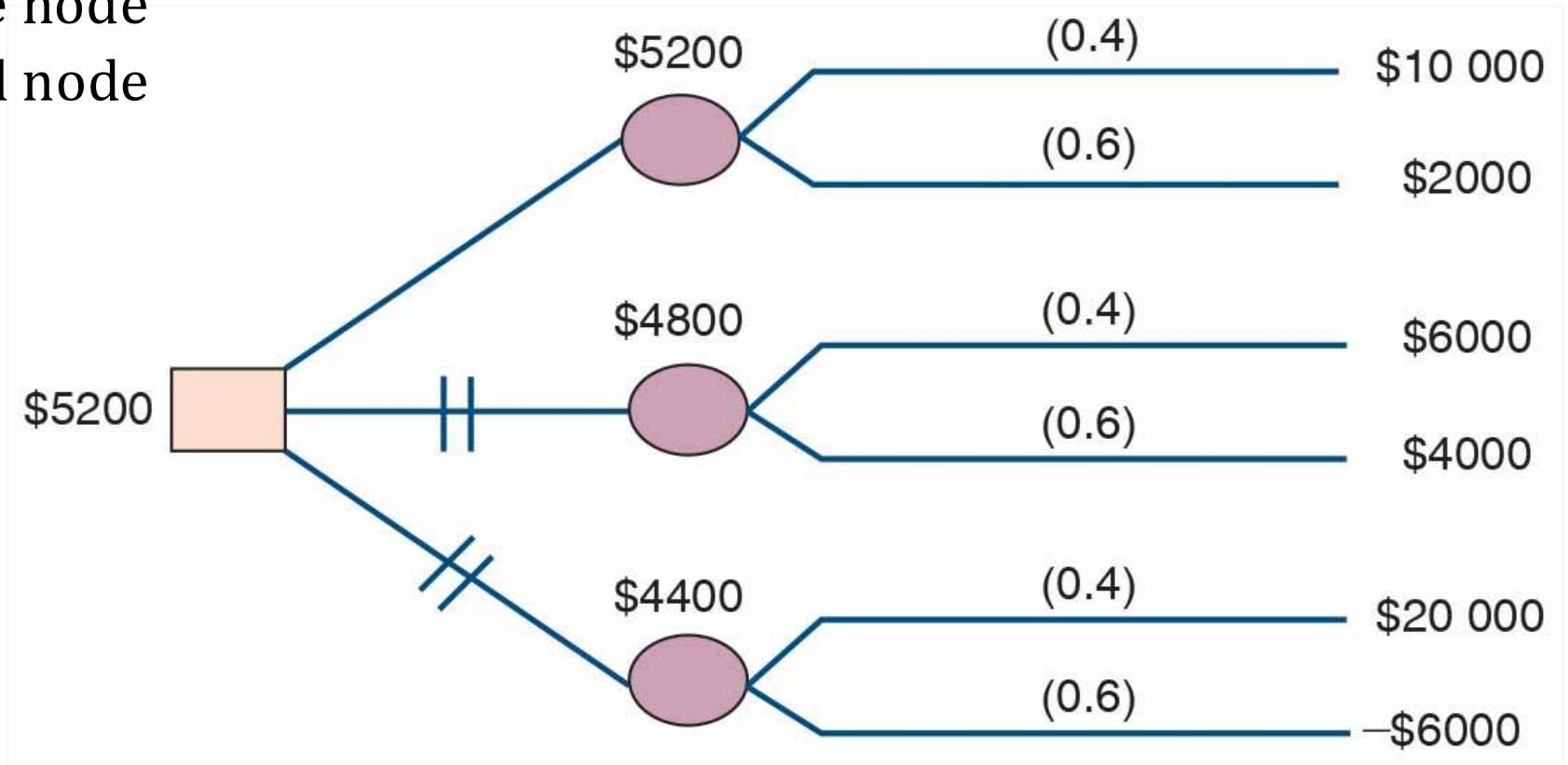
- Information in decision tables can be displayed as decision trees
- A decision tree is a graphic display of the decision process that indicates decision alternatives, states of nature and their respective probabilities, and payoffs for each combination of decision alternative and state of nature
- Appropriate for showing sequential decisions

Constructing a Decision Tree

1. Define the problem
2. Structure or draw the decision tree
3. Assign probabilities to the states of nature
4. Estimate payoffs for each possible combination of decision alternatives and states of nature
5. Solve the problem by working backward through the tree computing the EMV for each state-of-nature node

Decision Tree

- All decision trees contain:
  - Decision node
  - Outcome node
  - Terminal node



Decision Tree – Folding Back

- We analyze decision tree by folding back the tree:

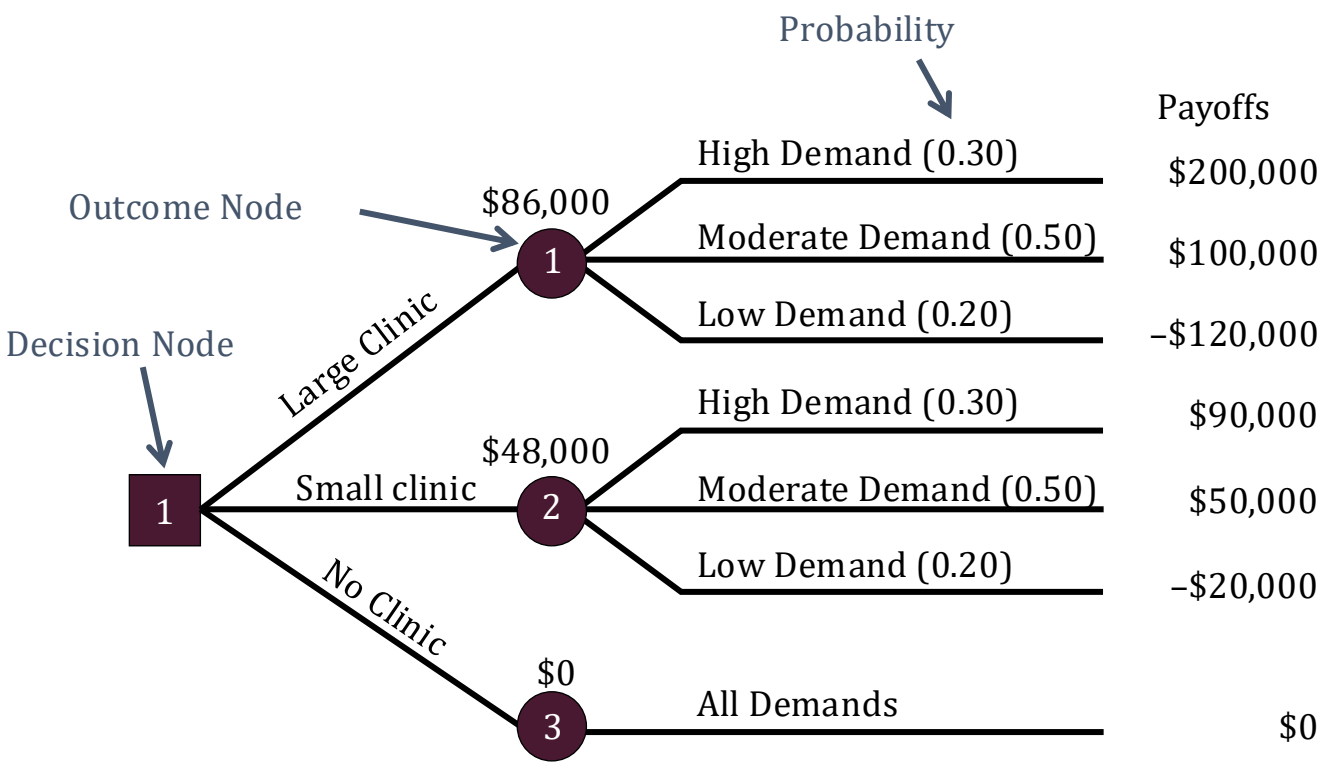
- At each outcome node
  - Calculate the expected monetary value (EMV)

- At each decision node
  - Select the alternative with the best expected monetary value (EMV)

Example – Decision Tree

Alternatives		Outcomes		
		High Demand	Moderate Demand	Low Demand
1	Build Large Clinic	200,000	100,000	-120,000
2	Build Small Clinic	90,000	50,000	-20,000
3	Do Nothing	0	0	0

Example – Decision Tree



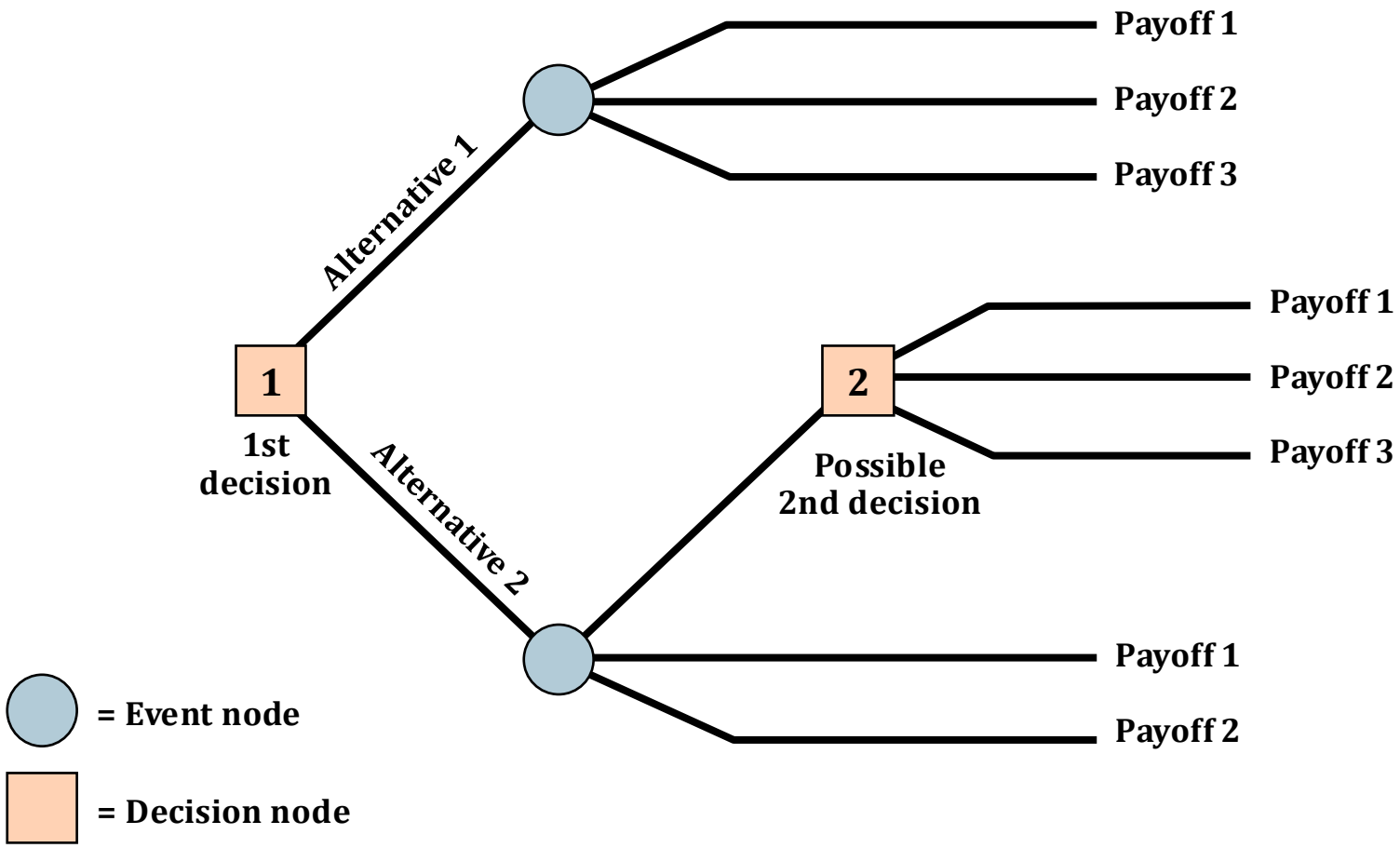
EMV 1 = 0.3($200,000) + 0.5($100,000) + 0.2(–$120,000) = $86,000

EMV 2 = 0.3($90,000) + 0.5($50,000) + 0.2(–$20,000) = $48,000

Decision Tree for Multistage Problems

- Problems with a sequence of alternatives and outcomes
- Decision → Outcome→ Decision → Outcome …
- It is possible that:
  - Decision → Decision
  - Outcome → Outcome

Multistage Decision Trees

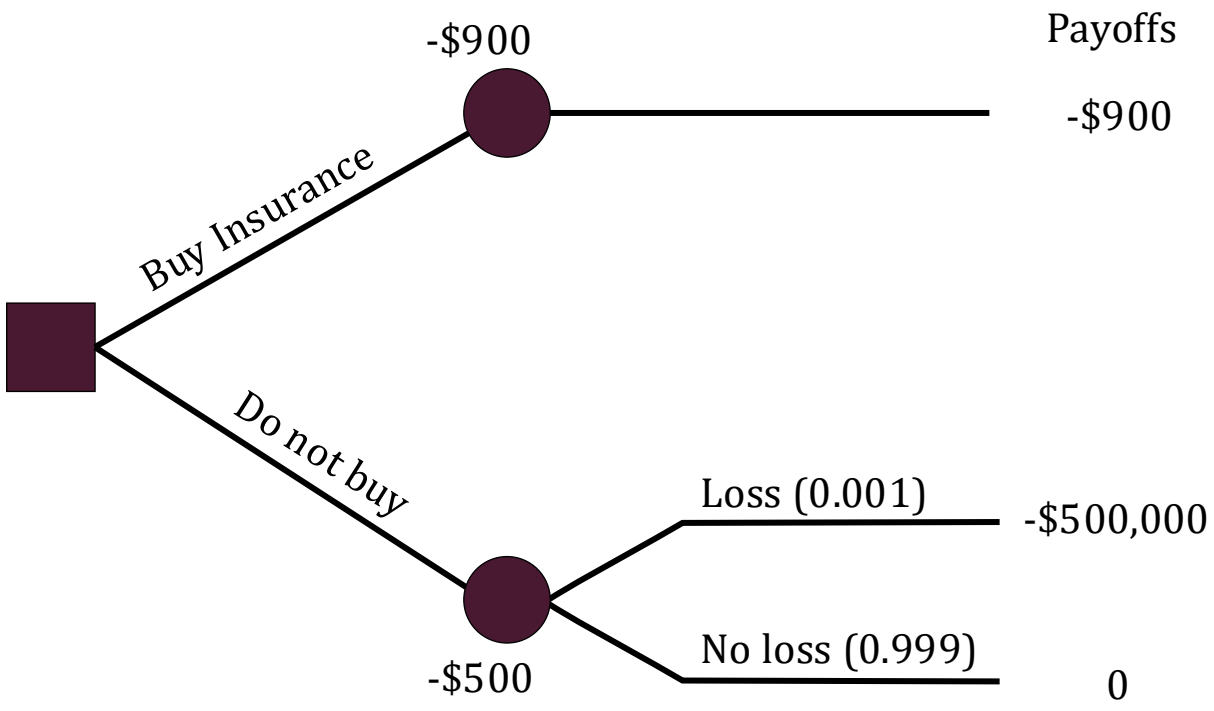


Decision Tree

- In decision trees, the assumption is that we always calculate EMV.
- But do we always do that in real life?
- Why some people are more willing to take risk than others? Can we use decision tree to model that?

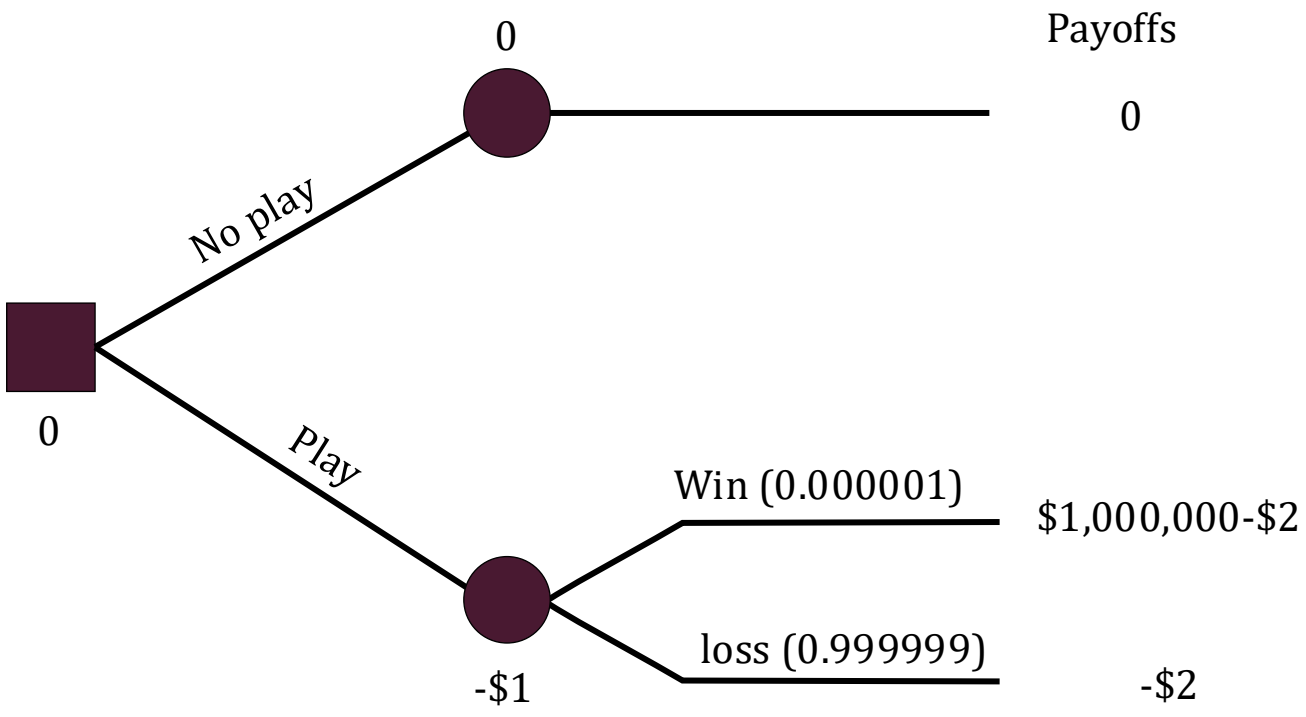
Home Insurance Example

- If we use EMV, we should not buy insurance.



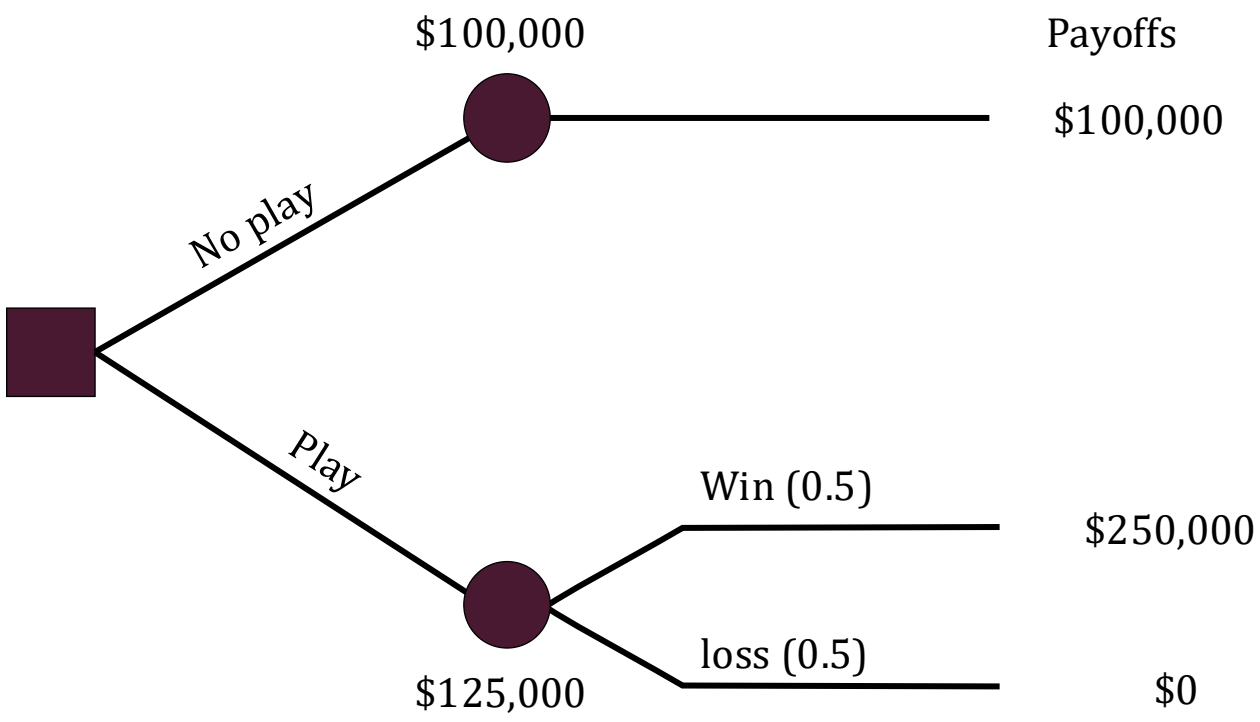
Lottery Example

- If we use EMV, we should not pay $2 for lottery ticket.



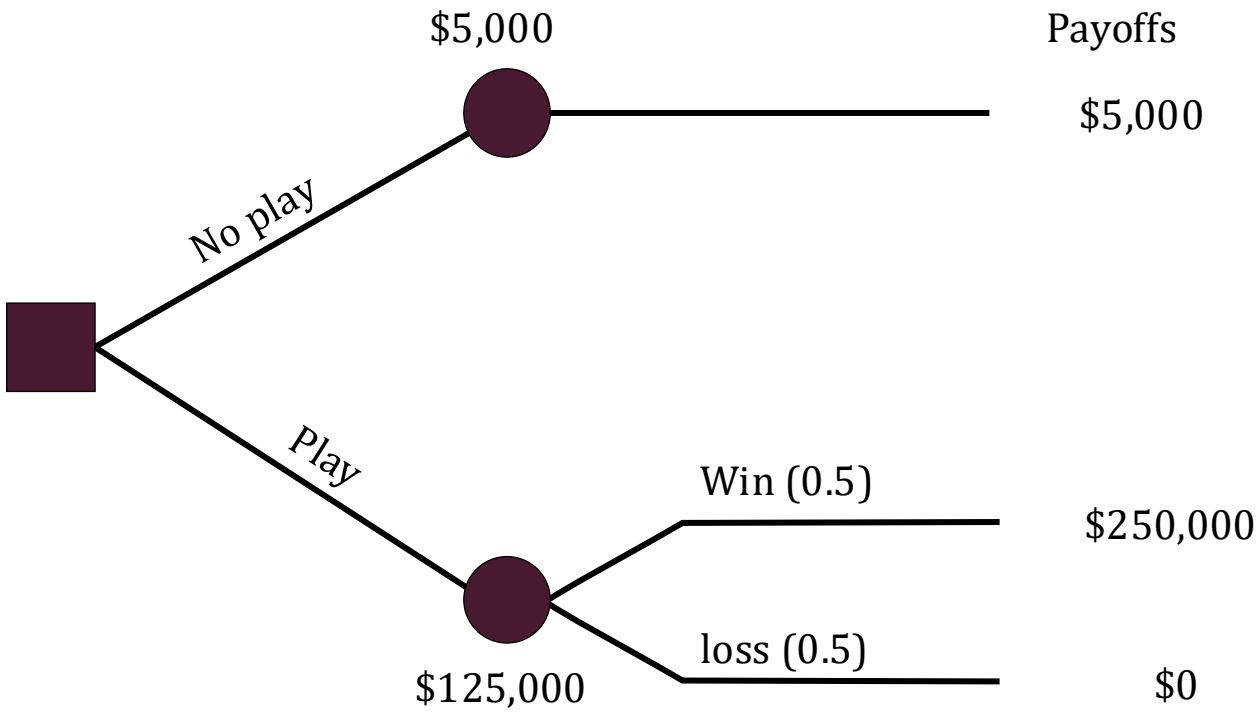
Game Show Example

- What do you do in this situation?



Game Show Example

- What do you do in this situation?



Utility Theory

- These examples showed that people (decision makers) have different attitudes toward risk when making decisions.

- We would like to incorporate a person's attitude toward risk, and a person's value for money in the decision modelling.

- Utility theory does this for us.

Utility Function

- Utility function converts a person's value for money and attitudes toward risk into a number <u>usually</u> in [0,1].
  - The worst payoff → 0
  - The best payoff → 1
- Each person has a different utility function.
- The utility function of a person can change over time.
- A person may have different utility functions in different situations:
  - For small monetary amounts, people accept more risks.
  - For large monetary amounts, people act more conservative.

Principals of Utility Theory

1. If alternative A is preferred to B, the utility of A is greater than the utility of B.

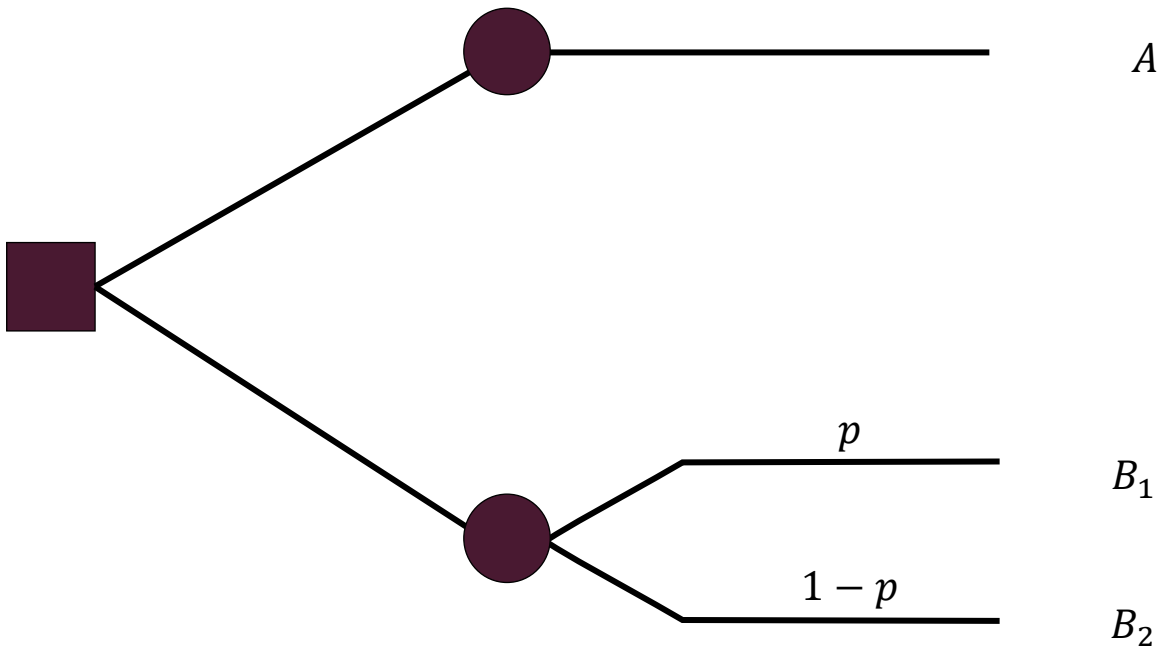
$$A \succ B \Leftrightarrow \quad U(A) > U(B)$$

Principals of Utility Theory

2. If decision maker is indifferent between
   a) A for certain;
   b) B with probability p for success and probability $1-p$ for failure,

   then:

$$U(A) = p.U(B_1) + (1-p).U(B_2)$$

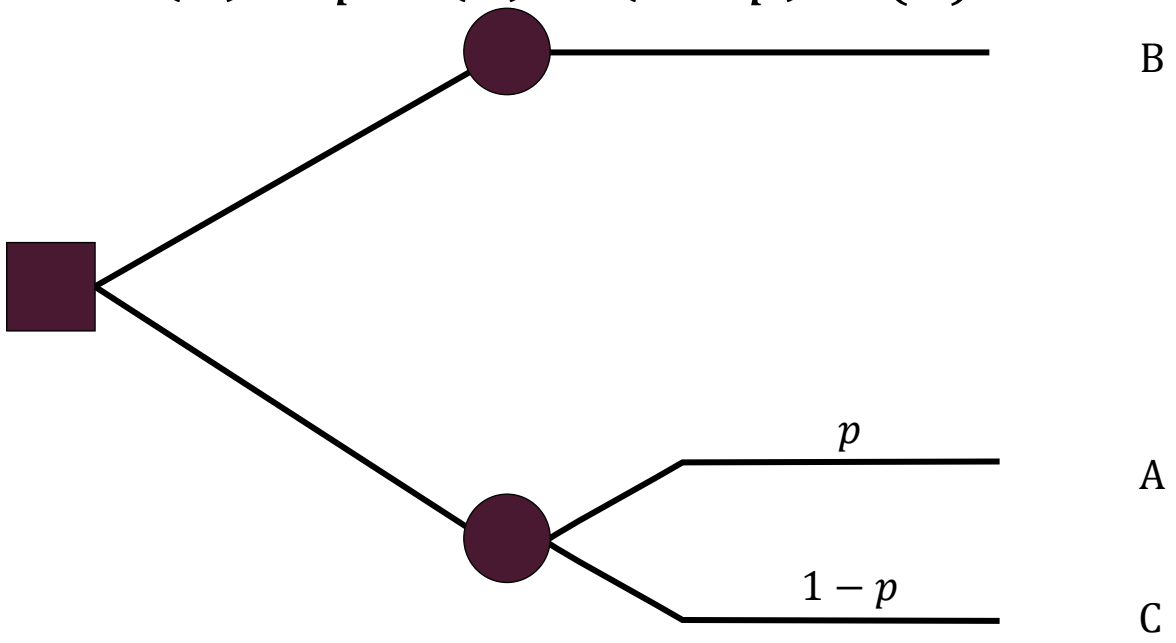


Principals of Utility Theory

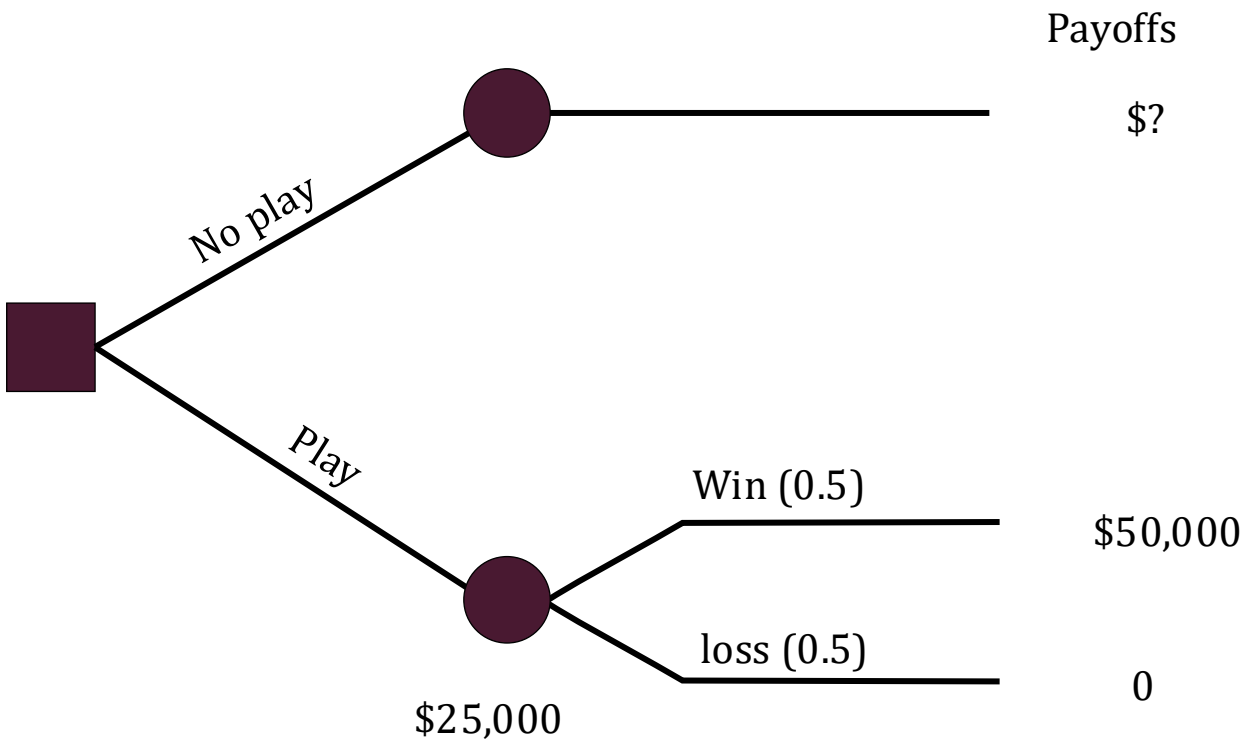
3. Certainty Equivalent:
   - Is the minimum guaranteed amount you are willing to accept to avoid the risk of playing.

If $A \succ B \succ C$, then:

$$U(B) = p.U(A) + (1-p).U(C)$$

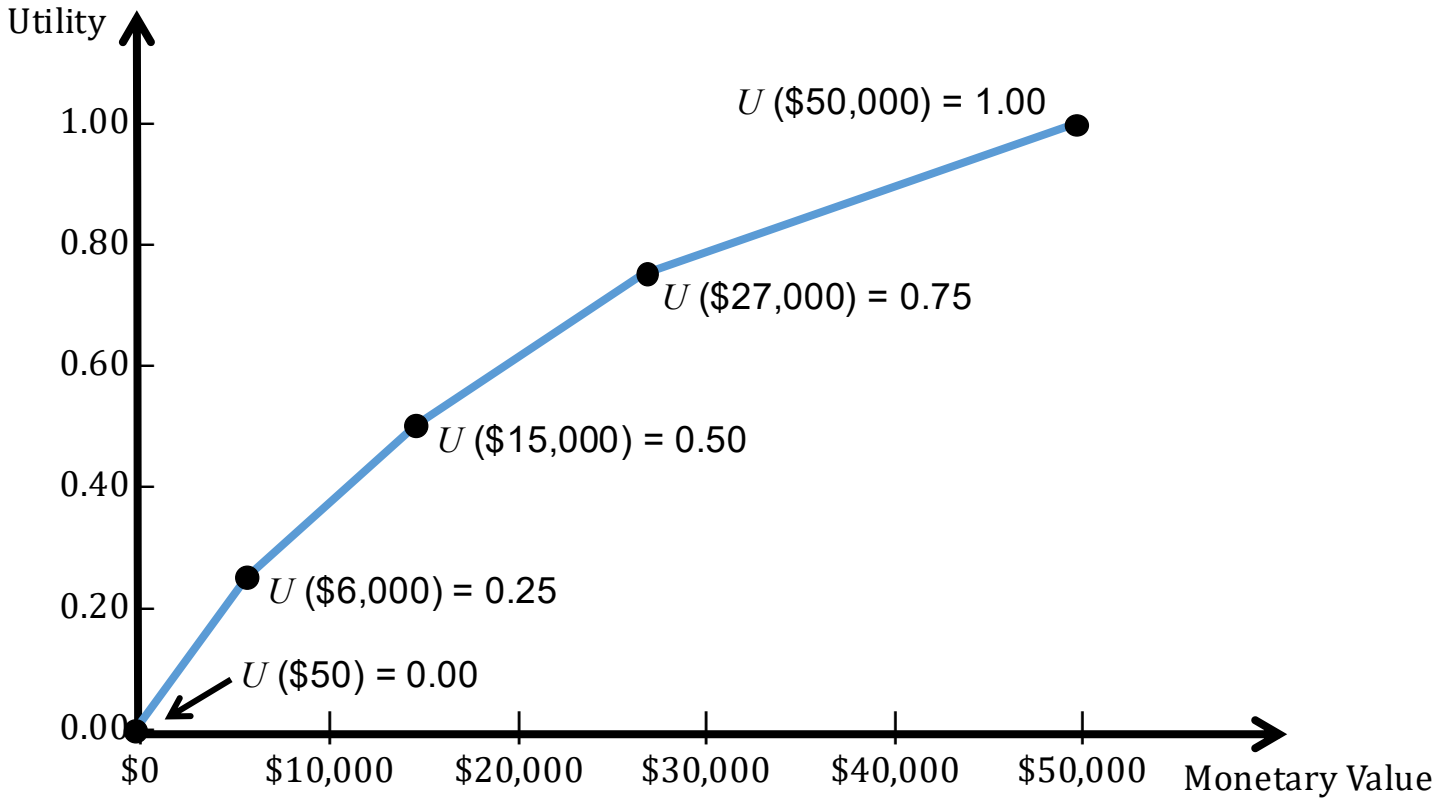


Gambling Example



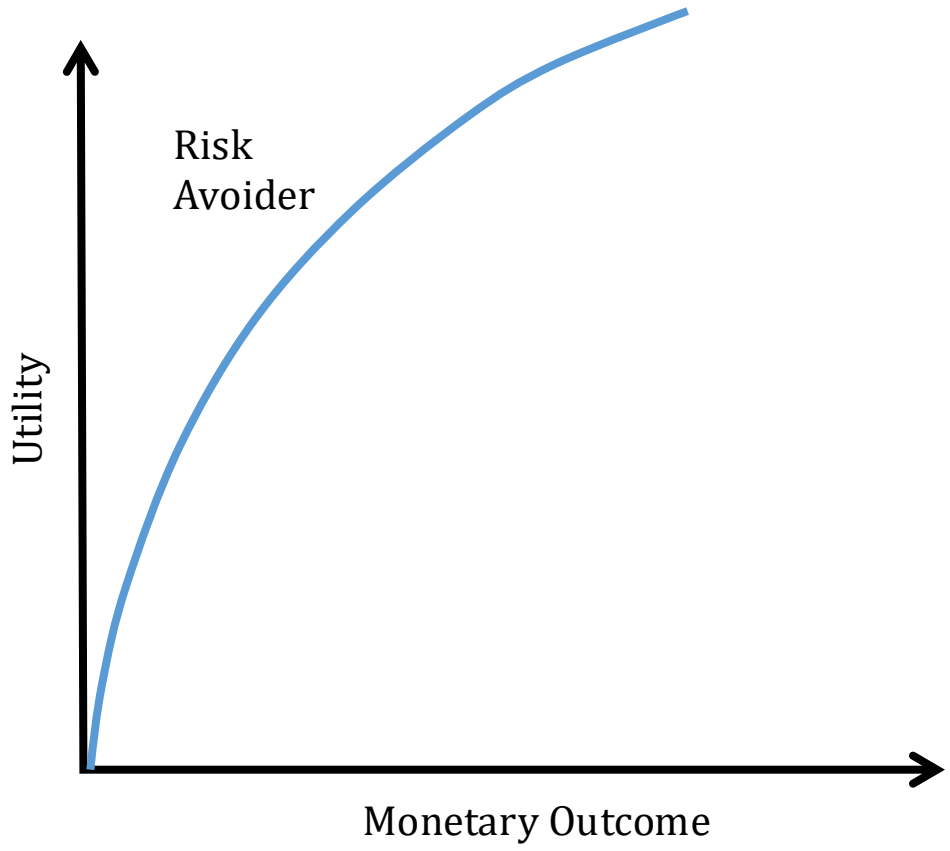
Utility Curve

- Using the mentioned principals, we can have utility curve, which plots utility values Vs. monetary values:



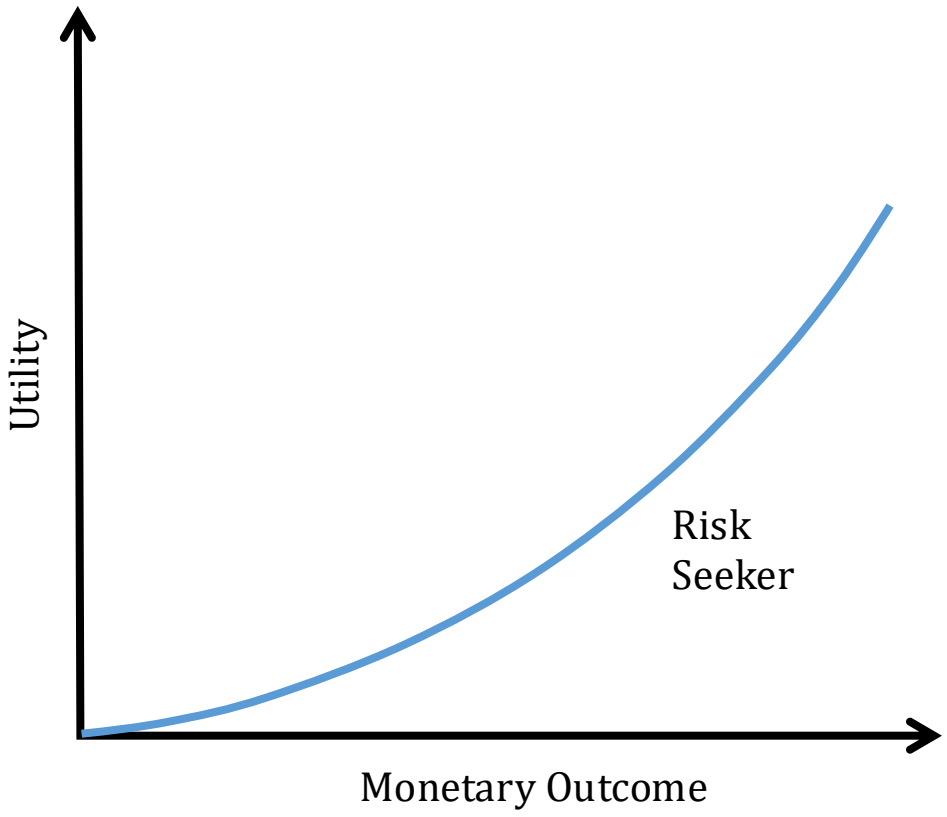
Risk Avoider (Risk Averse)

- Decreasing marginal utility for money



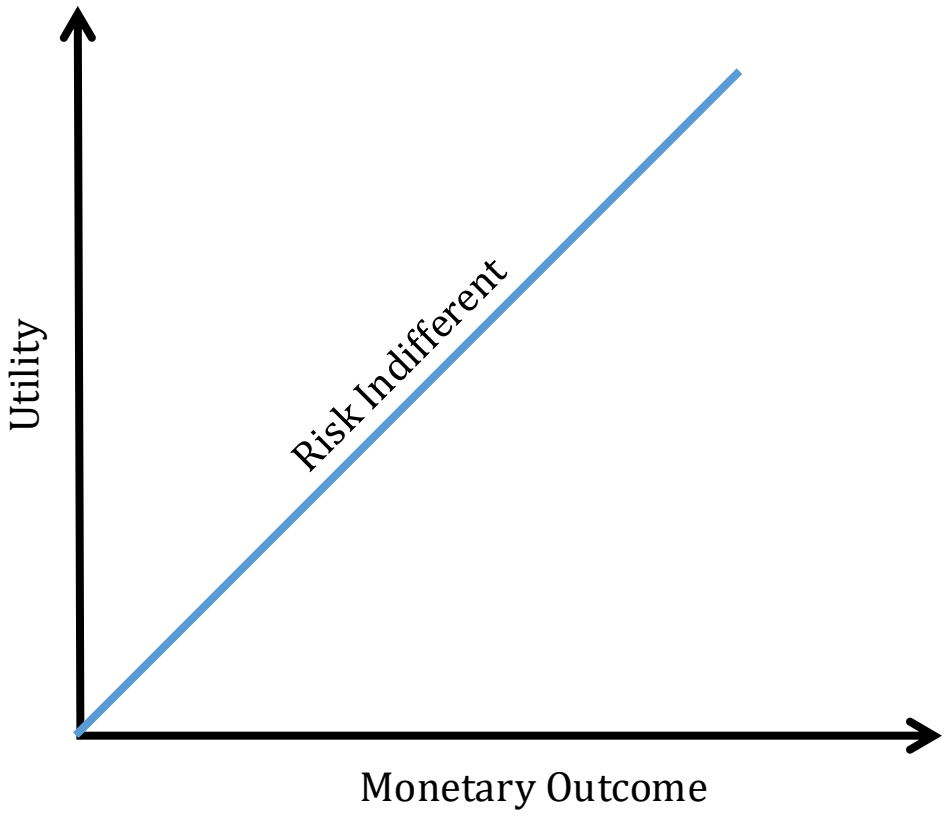
Risk Seeker

- Increasing marginal utility for money

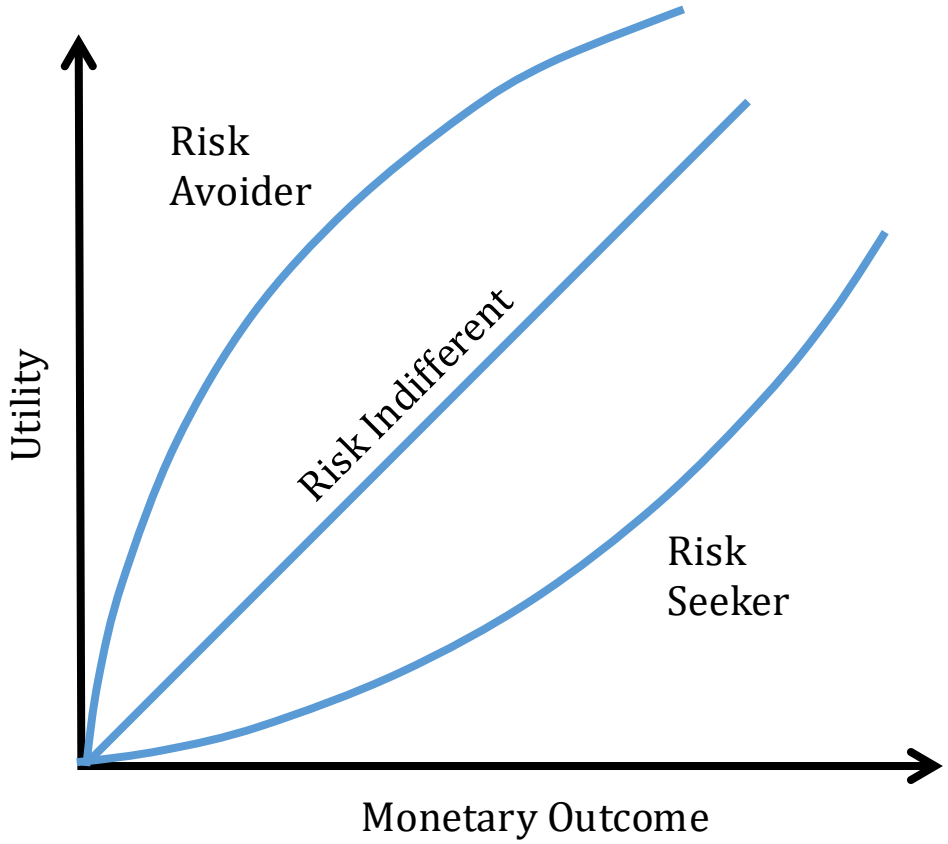


Risk Neutral

- Fixed marginal utility for money



Utility Curves



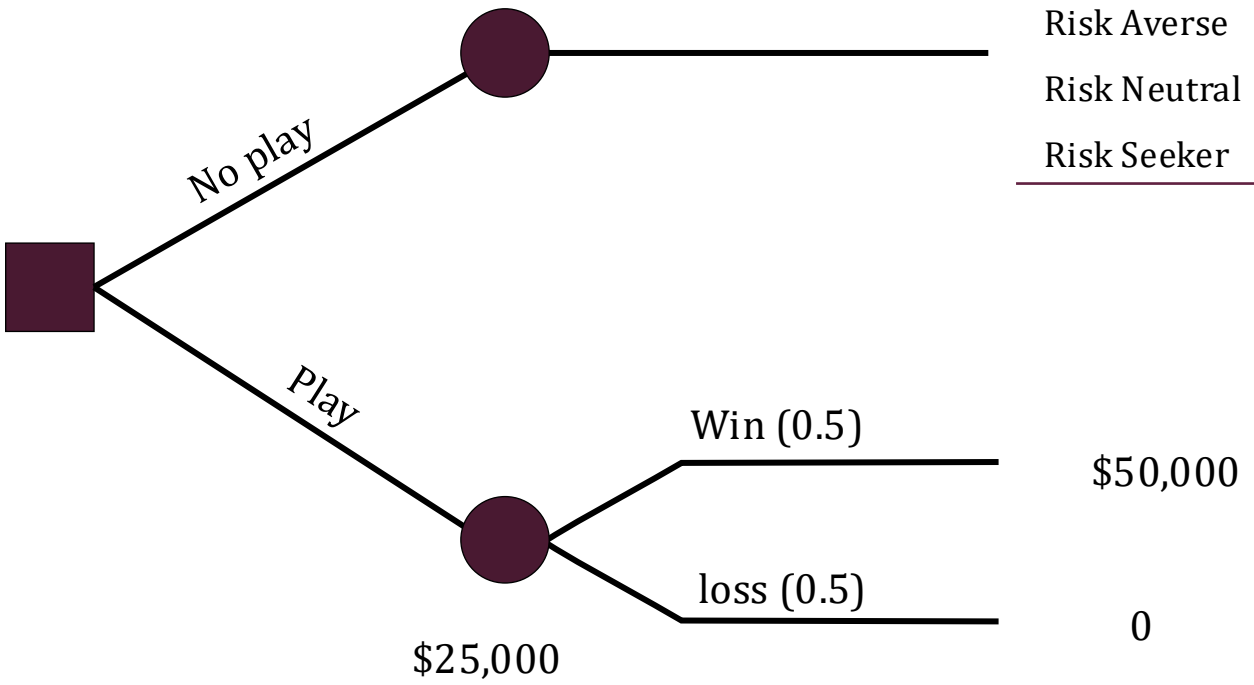
Risk Premium

- A risk averse person will give up larger amounts of money to avoid risk
- A risk seeker wants a certainty equivalent even greater than EMV to walk away from a play
- A risk neutral will decide based on EMV

$$\text{Risk Premium} = EMV - CE$$

$$\begin{aligned} \text{Risk Premium} > 0 \quad &\leftrightarrow \quad \text{Risk averse} \\ \text{Risk Premium} < 0 \quad &\leftrightarrow \quad \text{Risk seeker} \\ \text{Risk Premium} = 0 \quad &\leftrightarrow \quad \text{Risk neutral} \end{aligned}$$

Risk Premium - Gambling Example



	CE	Risk Premium
Risk Averse	$ 10,000	$ 15,000
Risk Neutral	$ 25,000	$ 0
Risk Seeker	$ 30,000	$ -5,000

Decision Making with Sampling (Experiments)

- Frequently, additional testing can be done to improve the preliminary estimate of the probabilities of the respective state of nature provided by the prior probabilities.

- These improved estimates are called **posterior probabilities**.

- Posterior Probabilities are essential in Decision Analysis, where we must weigh the value of acquiring new information (such as market research or medical tests) before making a final commitment.

Three Pillars of Bayes

- Prior Probability P(A): Our initial belief or estimate of the probability of an event before any new evidence is seen.

- Likelihood P(B|A): The probability of observing the new evidence given that the event A is true (often derived from historical accuracy or test specifications).

- Posterior Probability P(A|B): Our revised belief—the probability that event A is true after considering the new evidence B.

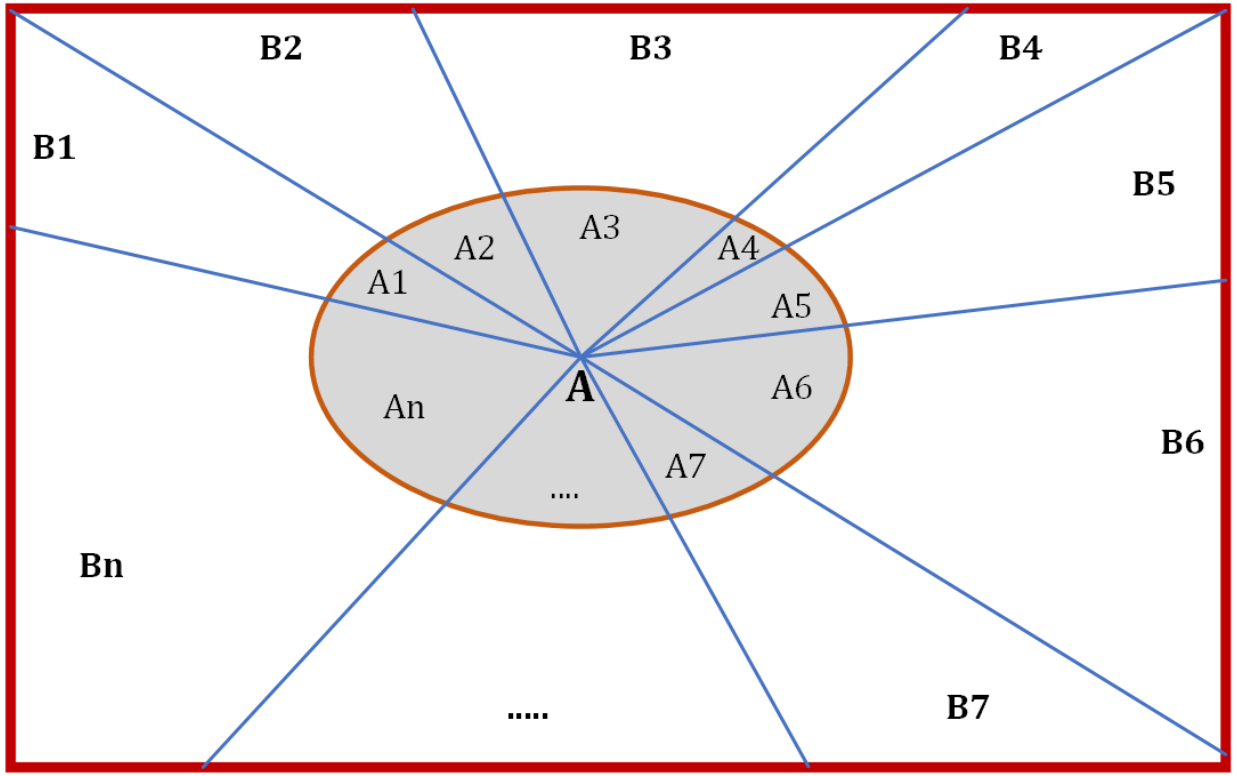
How to find Posterior Probabilities?

- We use Bayes Theorem to calculate them. First:

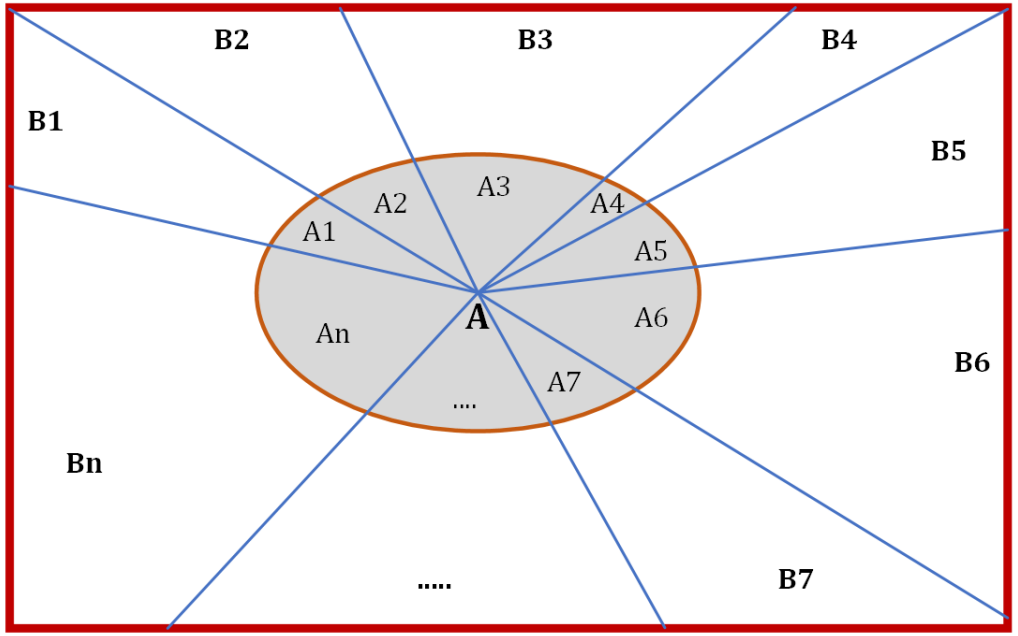
$$P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{P(A|B).P(B)}{P(A)}$$

Bayes Rules

$$P(A) = \sum_{i=1}^{n} P(A_i) = \sum_{i=1}^{n} P(A \cap B_i) = \sum_{i=1}^{n} P(A|B_i)P(B_i)$$



Bayes Rules

$$P(A) = \sum_{i=1}^{n} P(A_i) = \sum_{i=1}^{n} P(A \cap B_i) = \sum_{i=1}^{n} P(A|B_i)P(B_i)$$

$$P(B_k|A) = \frac{P(B_k \cap A)}{P(A)} = \frac{P(A|B_k)P(B_k)}{\sum_{i=1}^{n} P(A|B_i)P(B_i)}$$